User Manual

Online Learning Management System (LMS)

Gifts Academy

Project Title: Online Learning Management System

Developer: Kudakwashe Tawonga Tesa

Institution: BCS Higher Education Qualifications

Professional Diploma Project in IT

Contents

Introduction 3
1.1 Purpose of this Manual 3
1.2 System Background 3
1.3 System Objectives 4
1.4 Target Audience 4
2. System Overview 5
2.1 System Architecture 5
2.2 Key Features 5
2.3 User Roles 6
3. System Requirements 7
3.1 Hardware Requirements 7
3.2 Software Requirements 7
3.3 Network Requirements 8
4. Installation Guide 8
4.1 For End Users (No Installation Required) 8
5. Getting Started 9
5.1 Accessing the System 9
5.2 Creating an Account 9
5.3 Logging In 10
5.4 Logging Out 10
5.6 Navigation Overview 10
6. User Roles and Responsibilities 10
6.1 Student 10
6.2 Teacher 11
6.3 Parent 12
6.4 Administrator 12
7. Student Guide 13
7.1 Student Dashboard 13
7.2 Browsing and Enrolling in Courses 13

7.3 Accessing Course Content....14
7.4 Viewing Lessons....14
7.5 Submitting Assignments....15
7.7 Participating in Discussions....16
7.8 Profile Management....17
8. Teacher Guide....18
8.1 Teacher Dashboard....18
8.2 Creating and Managing Courses....18
8.3 Creating Lessons....19
8.4 Creating Assignments....20
8.8 Posting Announcements....21
10. Administrator Guide....22
10.1 Administrator Dashboard....22
10.2 Managing User Accounts....22
10.3 Course Management....23
10.3.1 Viewing All Courses....23
10.3.2 Assigning Teachers to Courses....23
10.4 Viewing Statistics....23
14. Conclusion....24
14.3 Support and Maintenance....24

Introduction

1.1 Purpose of this Manual

This user manual provides comprehensive documentation for the Online Learning Management System (LMS) developed for Gifts Academy. The manual was designed to assist all users including students, teachers, parents, and system admins on using the system's features and functionalities.

1.2 System Background

The Online Learning Management System was developed in response to the Zimbabwe government's national e-learning strategy (launched March 2021), which aims to improve access to quality education through digital solutions by 2025. Gifts Academy, like many schools in Zimbabwe, faced challenges with manual teaching methods including:

- Manual note-taking and distribution
- Disorganized homework sharing via WhatsApp
- Limited access to lessons for absent students
- Lack of centralized learning platform

This system addresses these challenges by providing a centralized, web-based platform for course management, content delivery, assignment and submission.

1.3 System Objectives

The LMS aims to:

1. Provide a centralized platform for teachers to upload lessons, assignments, and resources

2. Enable online assignment submission and grading

3. Ensure students can access lessons anytime, even when absent

4. Helps teacher to student interaction through announcements and discussion forums

5. Optimize for low-bandwidth usage and mobile devices

1.4 Target Audience

This manual is intended for:

- **Students**: Learners accessing course materials and submitting assignments
- **Teachers**: Teachers creating and managing course content
- **Parents**: Guardians monitoring student progress
- **Administrators**: System managers who oversee user accounts and system configuration

2. System Overview

2.1 System Architecture

The Online Learning Management System is a web-based application built using the following technologies:

Component	Technology	Version
Backend Framework	Django	5
Programming Language	Python	3.11
Database	SQLite	3
Frontend	HTML5, CSS3	
CSS Framework	Bootstrap 5	5.3.0

2.2 Key Features

2.2.1 User Management

- User registration with role selection Student, Teacher, Parent
- Secure authentication and password protection
- Profile management
- Role-based access control

2.2.2 Course Management

- Course creation and organization

- Course enrolment for students
- Teacher assignment to courses

2.2.3 Content Delivery

- Lesson creation
- File uploads (PDF, Word, images)
- Video URL embedding
- Downloadable notes

2.2.4 Assignment Management

- Assignment creation with due dates
- Online submission (file upload or text)
- Grading system
- Late submission detection

2.2.5 Communication

- Course announcements
- Lesson comments and replies
- Discussions

2.2.6 Progress Tracking

- Student dashboards with enrolled courses
- Teacher dashboards with pending grading
- Parent dashboards with student progress
- Admin dashboards with system statistics

2.3 User Roles

The system supports four distinct user roles:

Role	Description

Student	Learners who enrol in courses, access materials, submit assignments, and view grades
Teacher	Teachers who create courses, upload lessons, post assignments, and grade submissions
Parent	Parents who monitor their children's academic progress.
Administrator	System managers who oversee user accounts, approve registrations, and configure settings.

3. System Requirements

3.1 Hardware Requirements

For Users

Device	Minimum Requirement	Recommended
Computer	1 GHz processor, 1 GB RAM	2 GHz processor, 4 GB RAM
Tablet	Android/iOS, 1 GB RAM	2 GB RAM
Smartphone	Android 8 / iOS 12, 1 GB RAM	Android 11 / iOS 15, 3 GB RAM
Storage	100 MB free space	500 MB free space
Display	800 x 600 resolution	1366 x 768 or higher

3.2 Software Requirements

For Users

Software	Minimum Version	Recommended
Web Browser	Chrome 90, Firefox 88, Safari 14	Latest version
Operating System	Windows 10, macOS 11, Android 10, iOS 14	Latest version
Internet Connection	256 kbps	1 Mbps or higher
PDF Reader	Any PDF reader	Any PDF reader

3.3 Network Requirements

- Internet Connection: Stable connection recommended
- Firewall: Port 80 (HTTP) and 443 (HTTPS).
- Bandwidth: Optimized for low-bandwidth connections (as low as 256 kbps)

4. Installation Guide

4.1 For End Users (No Installation Required)

The Online Learning Management System is a web-based application. No installation is required for students, teachers, or parents. Users simply need:

1. A device with internet connection
2. A modern web browser
3. login credentials

To access the system:

1. Open your web browser
2. Navigate to the school's LMS URL (provided by the administrator)
3. Login with your credentials

5. Getting Started

5.1 Accessing the System

1. Open your web browser
2. Enter the LMS URL: https://online-lms-django-2.onrender.com or the address provided by your school
3. The login page will appear

5.2 Creating an Account

If you do not have an account:

1. Click the "Register" link below the login button
2. Complete the registration form:
   - **Username**: Choose a unique username
   - **Email**: Provide a valid email address
   - **Password**: Create a strong password (at least 8 characters)
   - **Confirm Password**: Re-enter your password
   - **User Type**: Select your role (Student, Teacher, or Parent)
3. Click "Register"

5.3 Logging In

1. Enter your Username and Password
2. Click "Login"
3. You will be redirected to your role-specific dashboard

5.4 Logging Out

To log out securely:

1. Click on your username in the top navigation bar
2. Select "Logout" from the dropdown menu
3. You will be returned to the login page

5.6 Navigation Overview

The system uses a consistent navigation structure:

- Top Navigation Bar: Contains the logo, main menu links, and user profile dropdown
- Sidebar (Dashboard): Quick links to frequently used sections
- Breadcrumbs: Shows your current location within the system
- Footer: Contains copyright information and links

6. User Roles and Responsibilities

6.1 Student

Primary Responsibilities:

- Enrol in courses
- Access lesson materials and notes
- Submit assignments by deadlines
- Participate in discussions
- Track personal progress and grades

Accessible Features:

- Student Dashboard
- Course browsing and enrolment
- Lesson viewing and note downloading
- Assignment submission
- Comment posting
- Grade viewing

6.2 Teacher

Primary Responsibilities:

- Create and manage courses
- Upload lesson materials
- Create assignments with deadlines
- Grade student submissions
- Post announcements
- Moderate discussions

Accessible Features:

- Teacher Dashboard
- Course creation and management
- Lesson creation
- Assignment creation
- Grading interface
- Announcement system

6.3 Parent

Primary Responsibilities:

- Monitor child's academic progress
- View grades and feedback
- Track assignment completion

Accessible Features:

- Parent Dashboard
- Student progress viewing
- Grade reports
- Course list viewing

6.4 Administrator

Primary Responsibilities:

- Manage user accounts
- Approve new registrations
- Link parents to students
- Monitor system usage
- Generate reports
- Configure system settings

Accessible Features:

- Admin Dashboard
- User management
- Course oversight
- System statistics
- Report generation

7. Student Guide

7.1 Student Dashboard

Dashboard Components:

Section	Description
Welcome Message	Personalized greeting
My Courses	Cards showing enrolled courses with quick access
Upcoming Assignments	List of assignments due soon
Recent Submissions	Recently submitted work with grades

7.2 Browsing and Enrolling in Courses

To browse available courses:

1. Click "Courses" in the top navigation menu
2. All available courses are displayed as cards
3. Each card shows:
    - Course title and code
    - Brief description
    - Teacher name
    - Number of enrolled students

To enrol in a course:

1. Find the desired course
2. Click the "Enrol Now" button
3. A confirmation message will appear
4. The course now appears in your dashboard under "My Courses"

7.3 Accessing Course Content

To view course materials:

1. From your dashboard, click "Go to Course" on any course card
2. The Course Detail page opens with three main sections:

Section	Content
Announcements	Latest news from your teacher
Lessons	Course content organized chronologically
Assignments	Tasks to complete with due dates

7.4 Viewing Lessons

To access lesson materials:

1. In the Course Detail page, locate the Lessons section
2. Lessons are displayed in an accordion format
3. Click on any lesson title to expand it

Each lesson contains:

- Lesson Title: The topic or title
- Content: Text explanation and instructions
- Notes File: Downloadable materials (PDF, Word, etc.)
- Video URL: Link to supplementary video content
- Comments Section: Discussion area for questions

To download notes:

1. Expand the lesson
2. Click the "Download Notes" button
3. The file will download to your device

7.5 Submitting Assignments

To view assignments:

1. In the Course Detail page, locate the Assignments section
2. Click "View Assignment" on any assignment

Assignment details include:

- Title and description
- Due date and time
- Total points possible

- Your submission status

To submit an assignment:

1. Click "Submit Assignment" button
2. Choose submission method:

    - Click "Choose File"
    - Select a file from your device
    - Supported formats: PDF, DOC, DOCX, TXT, JPG, PNG
    - Maximum file size: 10MB

3. Click "Submit Assignment"
4. A confirmation message appears

7.7 Participating in Discussions

To comment on a lesson:

1. Expand the lesson you want to discuss
2. Scroll to the Comments section
3. Type your comment in the text box
4. Click "Comment"

To reply to a comment:

1. Find the comment you want to reply to
2. Type your reply in the input box below that comment
3. Click "Reply"

Comment Guidelines:

- Be respectful to teachers and fellow students
- Stay on topic related to the lesson
- Use appropriate language
- Ask questions to clarify understanding

7.8 Profile Management

To view or edit your profile:

1. Click your username in the top navigation bar
2. Select "Profile" from the dropdown
3. View your account information
4. Update fields as needed:
    - Phone number
    - Address
    - Profile picture
5. Click "Save Changes"

8. Teacher Guide

8.1 Teacher Dashboard

Dashboard Components:

Section	Description
My Courses	All courses you teach with student counts
Pending Grading	Submissions awaiting your review
Assignment Overview	Summary of assignment status by course

8.2 Creating and Managing Courses

To create a new course:

1. From your dashboard, click "Create New Course"
2. Complete the course form:

Field	Description
Course Title	Full name of the course (e.g., "Mathematics Form 4")
Course Code	Unique identifier (e.g., "MATH-F4-01

Description	Overview of course content and objectives

3. Click "Create Course"

4. The course is now active and visible to students

To manage an existing course:

1. From your dashboard, click on any course card

2. The Course Detail page opens with management options:

- Add Lesson: Create new lesson content
- Add Assignment: Create new assignments
- Add Announcement: Post news to students
- View Roster: See enrolled students

8.3 Creating Lessons

To create a new lesson:

1. Go to your course page

2. Click "Add Lesson"

3. Complete the lesson form:

Field	**Description**
Lesson Title	Topic or title of the lesson
Content	Main lesson text
Notes File	Upload supporting materials (PDF, DOC, etc.)

Video URL	Optional link to YouTube or other video

4. Click "Save Lesson"

5. The lesson appears in your course immediately

Lesson Best Practices:

- Break content into manageable sections
- Include clear learning objectives
- Provide examples and illustrations
- Link to relevant resources
- Proofread before publishing

8.4 Creating Assignments

To create a new assignment:

1. Go to your course page

2. Click "Add Assignment"

3. Complete the assignment form:

Field	**Description**
Title	Name of the assignment
Description	Instructions, requirements, and expectations
Due Date	Date and time submission is required

4. Click "Create Assignment"

5. The assignment is now visible to enrolled students

8.8 Posting Announcements

To create an announcement:

1. Go to your course page
2. Click "Add Announcement"
3. Complete the announcement form:

Field	**Description**
Title	Brief, attention-grabbing headline
Content	Full announcement text

4. Click "Post Announcement"
5. All enrolled students see the announcement at the top of the course page

Announcement Uses:

- Class cancellations or schedule changes
- Exam reminders
- Important updates
- Motivational messages

10. Administrator Guide

10.1 Administrator Dashboard

Dashboard Components:

Section	Description
System Statistics	Total users, teachers, students, courses
Recent Users	Newly registered accounts
Recent Courses	Newly created courses

10.2 Managing User Accounts

To manage a specific user:

1. Find the user in the list
2. Click on their username
3. Available actions:

Action	Description
Activate/Deactivate	Enable or disable account access
Reset Password	Send password reset email

Change Role	Modify user type
Delete Account	Remove user (irreversible)

10.3 Course Management

10.3.1 Viewing All Courses

To see all courses:

1. Go to Course Management → All Courses

10.3.2 Assigning Teachers to Courses

To change course teacher:

1. Find the course in the list
2. Click "Change Teacher"
3. Select new teacher from dropdown
4. Click "Save"

10.4 Viewing Statistics

The admin dashboard displays:

- Total Users: All registered accounts

- Active Today: Users logged in last 24 hours
- Total Courses: All active courses
- Total Submissions: All assignment submissions

14. Conclusion

The LMS provides an efficient platform for managing teaching and learning activities. Users can easily interact with courses, assignments, and system features based on their roles.

14.3 Support and Maintenance

The system requires ongoing support to remain effective:

- Regular Backups: Daily database backups to prevent data loss
- Security Updates: Regular application of Django and dependency updates
- Feature Enhancements: Future updates based on user feedback

Document Version: 1.0

Prepared by: Kudakwashe Tawonga Tesa

For: Gifts Academy

This user manual is submitted as part of the project documentation for the BCS Higher Education Qualifications Diploma in IT.